

- Day 1:** 1. Each of the four cards shown below has a number on one side and a letter on the other side. What is the minimum number of cards that must be turned in order to prove the correctness of this statement: "If a card has a vowel on one side, then it has an even number on the other side?"

$$\boxed{A} \boxed{C} \boxed{4} \boxed{7}$$

Proof. We only need to check card $\boxed{A}$ and $\boxed{7}$. We check $\boxed{A}$ because if that is false, the claim doesn't hold. Also note the contrapositive of *If one side has vowel then there's even number on the other side* will be *If one side has odd number then there's a consonant on the other side*. So we check $\boxed{7}$ as well. The statement says nothing when one side is a consonant so we ignore that. The $\boxed{4}$ card will work either way so also ignored. $\square$

2. Is there a explicit bijection from $\mathbb{N} \rightarrow \mathbb{Q}$ or other way around?

Proof. Yes. Consider $S : \mathbb{N} \rightarrow \mathbb{Q}$ such that

$$S(x) = \begin{cases} \frac{\delta(x)}{2[x]-x+1} \\ 0, x = 0 \end{cases}$$

The sequence $S(0), S(S(0)), \dots$ gives all positive rationals uniquely. From this we can define a function $f : \mathbb{N} \rightarrow \mathbb{Q}$ as follows:

$$f(n) = \begin{cases} S^n(0), n \text{ is odd} \\ -S^n(0), n \text{ is even} \\ 0, n = 0 \end{cases}$$

$\square$

Day 2: Proposition(with proofs):

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$

Proof. $x \in \text{LHS} \implies x \in A \text{ and } x \notin B \cup C$

$\implies x \in A \text{ and } (x \notin B \text{ or } x \notin C)$

$\implies x \in A \setminus B \text{ and } x \in A \setminus C$

$\implies x \in (A \setminus B) \cap (A \setminus C)$

Thus, $\text{LHS} \subseteq \text{RHS}$.

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$\implies x \in A \setminus (B \cup C)$

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Hence, $\text{LHS} = \text{RHS}$. $\square$

2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

The argument via Venn diagram.

Day 3: Prove the well ordering principle of $\mathbb{N}$ using the principle of mathematical induction.

Proof. Let S_n be an arbitrary non-empty subset of $\mathbb{N}$ with n elements for some natural number n . Clearly if $n = 1, 2$ then S_n has a least element (If $n = 1$, that element is the minimal; if $n = 2$, we can compare the two elements and pick the smaller one). Assume S_n has a least element for $n = k \in \mathbb{N}_{>2}$. Then consider S_{k+1} . We know k of those elements are from a specific S_k . So, $S_{k+1} = S_k \cup \{\alpha\}$. Now simply compare the least element of S_k with α and pick the smaller one. Thus, S_{k+1} has a least element. Hence, by induction, every non-empty subset of $\mathbb{N}$ has a least element. $\square$

Day 4: